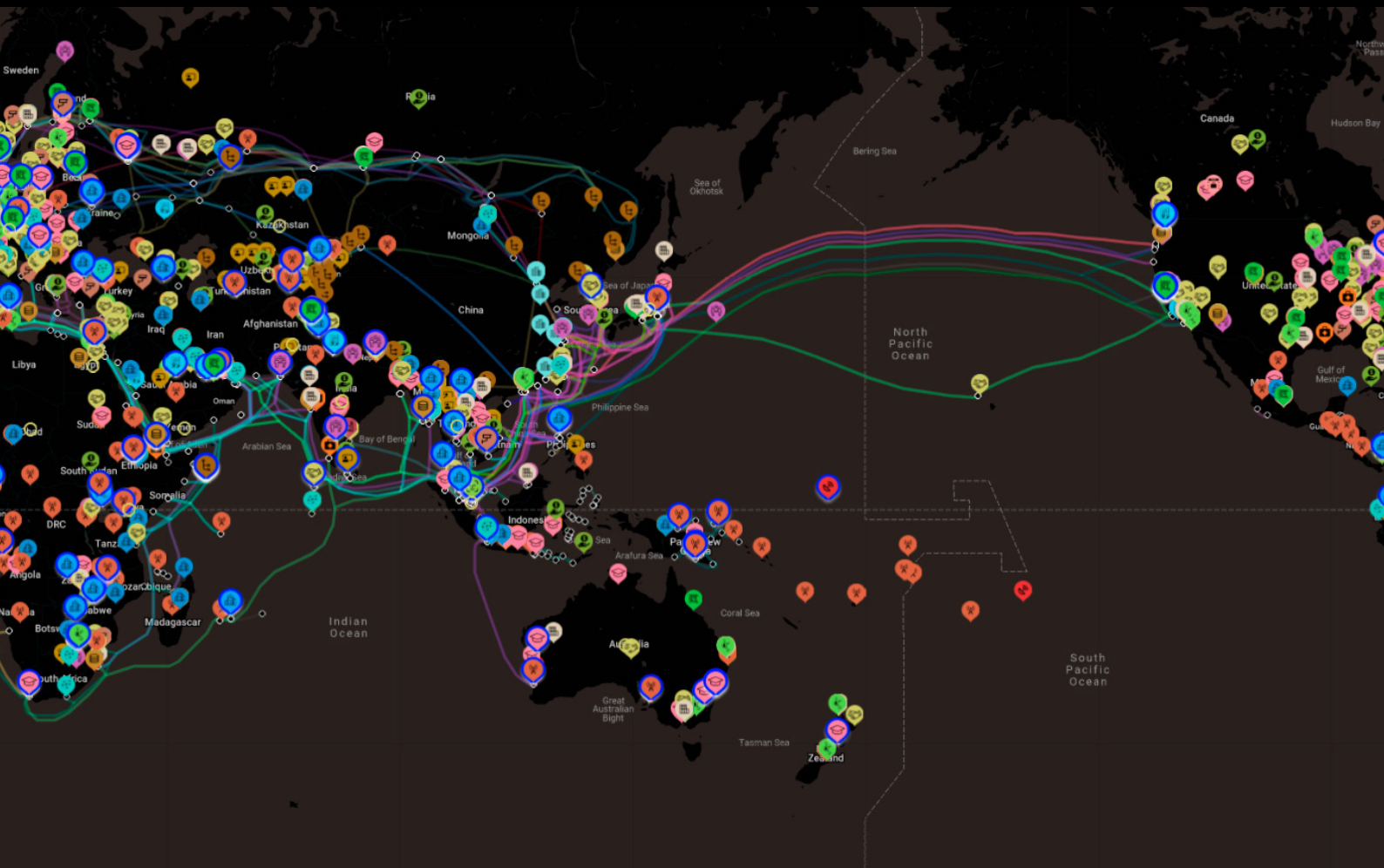


Mapping China's Technology Giants

Reining in China's technology giants

Fergus Ryan, Audrey Fritz and Daria Impiombato



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1. Introduction

Since the launch of ASPI ICPC's [Mapping China's Technology Giants project](#) in April 2019, the Chinese technology companies we canvassed have gone through a tumultuous period. While most were buoyed by the global Covid-19 pandemic, which stimulated demand for technology services around the world, many were buffeted by an unprecedented onslaught of sanctions from abroad, before being engulfed in a regulatory storm at home.

The environment in which the Chinese tech companies are operating has changed radically, as the pandemic sensitised multiple governments, multilateral groups and companies to their own critical supply-chain vulnerabilities. The lessons about national resilience learned from the pandemic are now being applied in many sectors, including the technology sector, where a trend towards decoupling China and the West was already well underway. As the geopolitical rivalry between the US and China has heightened, both sides increasingly see any reliance on the other for strategic commodities, such as rare-earth minerals and semiconductors, as dangerous vulnerabilities.

Supply-chain vulnerability has ignited work in Europe, North America and other regions to reduce dependence on China. Telecommunications companies such as [Huawei](#) and ZTE that are deemed 'high risk' by multiple countries are increasingly finding themselves locked out of developed markets. Amid the trade war between the US and China, which began in 2018, the Trump administration unleashed a relentless series of actions targeting Chinese companies in an effort to slow their advance. That onslaught has further convinced China's leadership to redouble its efforts to dominate the commanding heights of technology as a source of strategic and economic power.

Among the measures meted out by the Trump administration were limits on investment by Chinese technology companies,¹ blocks on the operations of Huawei and other Chinese telecom companies in the US,² pressure on other countries to block Huawei's operations,³ new export control regulations,⁴ tariffs on products benefiting from Beijing's 'Made in China 2025' program⁵ and an attempt to ban [ByteDance's](#) TikTok and [Tencent's](#) WeChat apps.⁶ The effects of the actions have been uneven—dealing a major blow to Huawei, for example, while barely touching the major Chinese internet firms' businesses.

For China's leadership, the twin crises of the Covid-19 pandemic and the growing China–US strategic and technological competition highlighted the country's need to achieve its long-held goal of 'technological self-reliance'.⁷ The US's ability to cut off China's technology companies' access to semiconductors, in particular, is seen by leaders from Xi Jinping down as an unacceptable 'choke point' holding back China's progress.⁸ The 14th Five-Year Plan, unveiled in March 2021, reflected the Chinese Communist Party's (CCP) sense of urgency. For the first time, it described technological innovation as a matter of national security, not just economic development.⁹

The now 27 Chinese technology firms that we cover on our Mapping China's Technology Giants project ('our map') span sectors including [biotechnology](#), [surveillance](#), [artificial intelligence](#) (AI), e-commerce, finance, entertainment and [telecommunications](#). All of them are set to play a key role in the coming years as Beijing ramps up major investments in strategic technologies such as 5G telecommunications, quantum computing and AI. Both state-owned and private businesses are being mobilised in a 'whole

country’ approach to reduce reliance on foreign technologies and seek breakthroughs in strategic science and technology projects.¹⁰ Beijing’s new goal is to increase R&D investment by 7% each year.¹¹ Already, several of the companies featured on our map, including [SenseTime](#), Huawei, ZTE, [Megvii](#), [YITU](#), [CloudWalk](#), [Baidu](#), [Alibaba](#), Tencent and China’s three major telecommunications companies, have been recruited into a US\$2 trillion ‘new infrastructure’ plan.¹²

Pushback on China’s technology giants didn’t just come from Washington, however; it also came from the CCP. Chinese regulators used the Covid-19 pandemic as an opportunity to tighten supervision over the companies, which had grown into behemoths with relatively light regulatory oversight in the past decade.¹³ The escalating geopolitical tensions with the US and the ensuing US–China trade war contributed to a government campaign to rein in Alibaba’s fintech affiliate [Ant Group](#), as the Chinese state sought to head off risks in the banking system amid concerns that the stand-off with Washington could precipitate a financial crisis.¹⁴ Those concerns culminated in the abrupt cancellation of the company’s initial public offering (IPO), which was set to be the world’s largest ever, just two days before its launch in Shanghai and Hong Kong in late 2020.¹⁵

Since then, the CCP’s efforts to tighten state control over China’s internet companies have widened. In April 2021, Chinese e-commerce leader Alibaba Group was hit with a record US\$2.81 billion antimonopoly fine, equivalent to around 4% of its 2019 domestic sales.¹⁶ A string of high-level resignations has followed as the government continues to seek to weaken the central authority of all the leaders of the major tech companies.¹⁷ China’s regulators, tasked with ‘tackling monopolies’ and ‘preventing disordered capital expansion’, have set their sights on a fundamental restructuring of the country’s biggest tech companies to ensure that they remain focused on technological innovation and align themselves even more closely with the strategic goals of the CCP.¹⁸

2. Covid-19

The Covid-19 pandemic has had a profound effect on the world economy. The International Monetary Fund estimates the global economy shrank by 4.4% in 2020, compared to a contraction of 0.1% in 2009 during the global financial crisis.¹⁹ China was no outlier in the first quarter of 2020, when its economy shrank by 6.8% in the first such contraction in at least 40 years.²⁰ Yet, amid the turmoil, technology giants—particularly in the US and China—provided a rare bright spot as they seized the opportunity to expand aggressively.

As reliance on digital products grew during the pandemic, demand for US and Chinese technology giants' products and services surged. The combined revenue of the largest US tech companies—Apple, Microsoft, Amazon, Google-parent Alphabet and Facebook—grew by a fifth to US\$1.1 trillion, while their combined market capitalisation grew by half during 2020 to US\$8 trillion.²¹ As of May 2021, the 27 companies we cover on our map had a combined market capitalisation of more than US\$2.2 trillion, ranking them, in estimated nominal GDP terms, as equivalent to the world's eighth largest economy, after France.²² Only three of the companies on our map—Huawei, Megvii²³ and CloudWalk—experienced slowing year-on-year revenue growth.

Some of China's internet companies, including Tencent, Alibaba, ByteDance, Huawei and biotechnology company BGI, attempted to turn the crisis into a public relations opportunity by providing financial or material assistance to countries struggling to control the Covid-19 pandemic (Figure 1). To take one example: Tencent's Covid-19 donations from its US\$100 million Covid-19 fund included medical equipment to sporting teams such as Football Club Barcelona²⁴ and the New England Patriots²⁵, cities such as Nashville (US)²⁶, countries such as Ethiopia²⁷, hospitals in Los Angeles (US)²⁸ and Karachi (Pakistan)²⁹, and the World Health Organization's Covid-19 Solidarity Response Fund.³⁰

Figure 1: China's technology giants' overseas donations



The Mapping China's Technology Giants project currently counts a total of more than 130 donations by all tracked companies combined. Over eighty of those donations are Covid-19 monetary and medical donations from ByteDance, Tencent and Alibaba.

Tencent's largesse was possible due to its oversized success. Supercharged by the pandemic, the company was able to exploit falling valuations to scoop up Norwegian game developer Funcom, take a stake in German developer Yager and make multiple investments in fintech start-ups, mainly in Europe and the US. The company currently sits on a portfolio worth roughly a quarter of a trillion dollars.³¹

As Chinese consumers ensconced themselves at home, Tencent's music and video service subscriber numbers swelled to 43 million and 112 million, respectively, growing by 50% and 26% from June 2019 to 2020.³² WeChat, the company's ubiquitous social media app, ballooned to over 1.2 billion users in the first quarter of 2020, up by more than 8% from 2019, as Tencent worked in collaboration with the Chinese Government's National Development and Reform Commission to create the WeChat Health Code app used to verify people's exposure to Covid-19.³³ Tencent's profit for the whole of 2020 stood at US\$25.1 billion (¥159.8 billion), a year-on-year increase of 71%.³⁴ At the time of writing, Tencent's market capitalisation is around US\$800 billion, making it China's most valuable company.

Despite 13 companies on our map having been added to the US Government's Entity List (see box below) and facing challenges while operating during the pandemic, many continued to report strong growth throughout 2020.

The Entity List

The US Department of Commerce's Entity List was created in 1997 to address risks related to the proliferation of weapons of mass destruction. The US Government has since expanded its basis for adding entities to the list to include countering Chinese military activity, countering spying and addressing human rights concerns.³⁵ Companies placed on the Entity List are banned from buying parts and components from US companies without government approval.

BGI, for example, saw its profits surge as Covid-19 spread around the world, despite the addition of two of its subsidiaries to the Entity List in July 2020. As of August 2020, BGI had already sold 35 million Covid-19 rapid-testing kits to 180 countries and built 58 labs in 18 countries (Figure 2).³⁶ Due to its rapidly expanding global presence, the company experienced a net profit surge of 653% during 2020, and the value of its shares climbed by 87%.³⁷ BGI's operating income in the North American market even increased by 556.23%, making up 9.91% of the company's total operating income in 2020.³⁸ By March 2021, BGI's market capitalisation on the Shanghai stock exchange had jumped to US\$7.9 billion (¥50.83 billion), up from its March 2020 market capitalisation of US\$5.26 billion (¥33.86 billion).³⁹

A world map showing the geographical distribution of 100 countries. The map is color-coded by continent: North America (light blue), South America (light green), Europe (light orange), Africa (light yellow), Asia (light pink), and Australia (light purple). The map includes labels for major countries and oceans. Icons are placed on the map to represent different types of organizations, such as NGOs, governments, academia, and the private sector. The icons are color-coded: blue for NGOs, green for governments, red for academia, and yellow for the private sector. The map shows a high density of icons in Europe and North America, with fewer icons in Africa and South America.

The [Mapping China's Technology Giants project](#) currently counts more than **100 points of presence for BGI** overseas, including commercial partnerships, Covid-19-related donations, investments, joint ventures, memorandums of understanding, overseas offices, research partnerships and subsidiaries.

WuXi AppTech Group is another biotech company that experienced growth during Covid-19, increasing its market capitalisation by 130%.⁴⁰ Since the beginning of the pandemic, WuXi has been involved in the research and production of antibody treatments for Covid-19, and in January 2021 announced its plans to begin producing vaccine components for British–Swedish pharmaceutical company AstraZeneca at WuXi’s manufacturing facility in Germany.⁴¹

Three of our mapped internet companies were responsible for donating notable sums of money globally in the fight to combat Covid-19. ByteDance, Tencent and Alibaba ranked in the world's top Covid-19 corporate financial donors, donating close to US\$436 million, US\$173 million and US\$144 million, respectively.⁴² Those sums fall behind donations from only two leading US technology companies: Google and Cisco donated US\$1.3 billion and US\$226 million, respectively.⁴³

The three Chinese companies also experienced significant growth in 2020:

- ByteDance's revenue more than doubled despite the challenges that its subsidiary TikTok faced, including a ban from the Indian market and attempts by the Trump administration to force TikTok's sale to an American owner.⁴⁴
- Similarly, Alibaba has been referred to as 'one of China's biggest corporate winners of the coronavirus crisis', as the company's online traffic skyrocketed in 2020 and the Chinese Government increased its reliance on Alibaba's cloud services in response to the pandemic.⁴⁵
- Ant Group, which is an affiliate of Alibaba, was essential in China's initial Covid-19 response. Early in the pandemic, the company assisted the Chinese Government in developing and implementing the Alipay Health Code to facilitate contact tracing.⁴⁶ Ant's small-business lending platforms accumulated a US\$300 billion credit balance, and its wealth management platform facilitated US\$590 billion worth of investments.⁴⁷

Similarly, Hikvision, Uniview, SenseTime,⁴⁸ iFlytek (Figure 3),⁴⁹ DJI, Meiya Pico⁵⁰ and Ping An Technology⁵¹—a collection of surveillance, AI and technology companies—grew by developing technology used in response to Covid-19. Many of those technologies include temperature-screening products and contact-tracing systems. SenseTime claimed it has improved its facial-recognition algorithm to identify individuals wearing masks using just the person's visible facial features.⁵²

Figure 3: iFlytek's Covid-19 impact

	In late January 2020, iFlytek donated medical equipment, supplies and educational technology to help Wuhan fight the coronavirus epidemic. iFlytek's AI medical imaging system was also used as part of the official response coordinated by the Chinese Academy of Sciences. In addition to its AI technology, iFlytek donated 'over 12,000 respirators, 3,000 sets of protective suits, and more than 13,000 goggles to hospitals in Hubei Province', which was the epicentre of the outbreak in China. iFlytek has also provided technical support for 'cloud classroom' initiatives by local governments in China. Binzhou's Bureau of Education, along with iFlytek, established a platform enabling students to log in with uniform accounts, view the weekly curriculum and arrange proper daily learning plans. iFlytek has partnered with South Korean technology company Hancom Group to launch Accufly.AI in South Korea. Accufly.AI is an AI outbound calling system and is to be used to assist the South Korean Government with contact tracing communication. Despite being on the US's Entity List, the company received an export licence from the US Department of Commerce to purchase medical supplies to help combat Covid-19.
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Source: This is an extract from one of our 'Thematic snapshots' on the Mapping China's Technology Giants project website (under 'Analysis'), [online](#).

Surveillance company Hikvision's revenue initially fell in the first quarter of 2020, but rebounded in the second quarter due to the company's overseas revenue growth from its 'fever cameras'.⁵³ Uniview followed a similar pattern, first experiencing a sales and profit slowdown in the first half of 2020 and then recovering by the end of the year due to strong overseas growth in temperature-screening products, according to our map (Figure 4).⁵⁴

Figure 4: Overseas expansion by Hikvision, Dahua and Uniview during the Covid-19 pandemic



The [Mapping China's Technology Giants project](#) depicts the overseas expansion of Hikvision, Dahua and Uniview as overseas demand for their temperature-screening products increased during the Covid-19 pandemic. The map contains [65 data points](#) of overseas presence relating to Covid-19 for the three companies, including donations, commercial partnerships and surveillance equipment.

Drones manufactured by technology company DJI proved useful in helping counter the spread of Covid-19. The company sold drones to countries, including France, Norway, Italy, the Philippines, Spain and Indonesia, and 22 states in the US to disinfect public areas and to patrol streets.⁵⁵

Although China's economic growth slowed to 2.3% by the end of 2020, its economy emerged as the only major economy expected to have grown in 2020 as a result of the pandemic.⁵⁶ China's digital economy, in particular, was positively affected by Covid-19, expanding by 9.7% from 2019.⁵⁷ While China's economic recovery had a head start, the International Monetary Fund expects the global economy to recover and grow by 6.1% in 2021, estimating 5.1% growth for advanced economies and 6.7% growth for developing economies.⁵⁸

Despite external pressures amid tense US–China relations, Covid-19 provided the technology giants on our map with an opportunity to expand both domestically and overseas. High-profile donations of personal protective equipment from the tech giants helped to burnish their brands as well as deflect criticism of the Chinese state's cover-up of the Covid-19 outbreak in its early days. China's tech giants may have received a short-term boost from the pandemic, but over the longer term their prospects are less certain as many countries begin to address their dependence on China in critical sectors.⁵⁹ As those countries make changes to reduce their reliance on China, the overseas growth that Chinese tech companies have experienced may slow.

3. US–China tech tensions

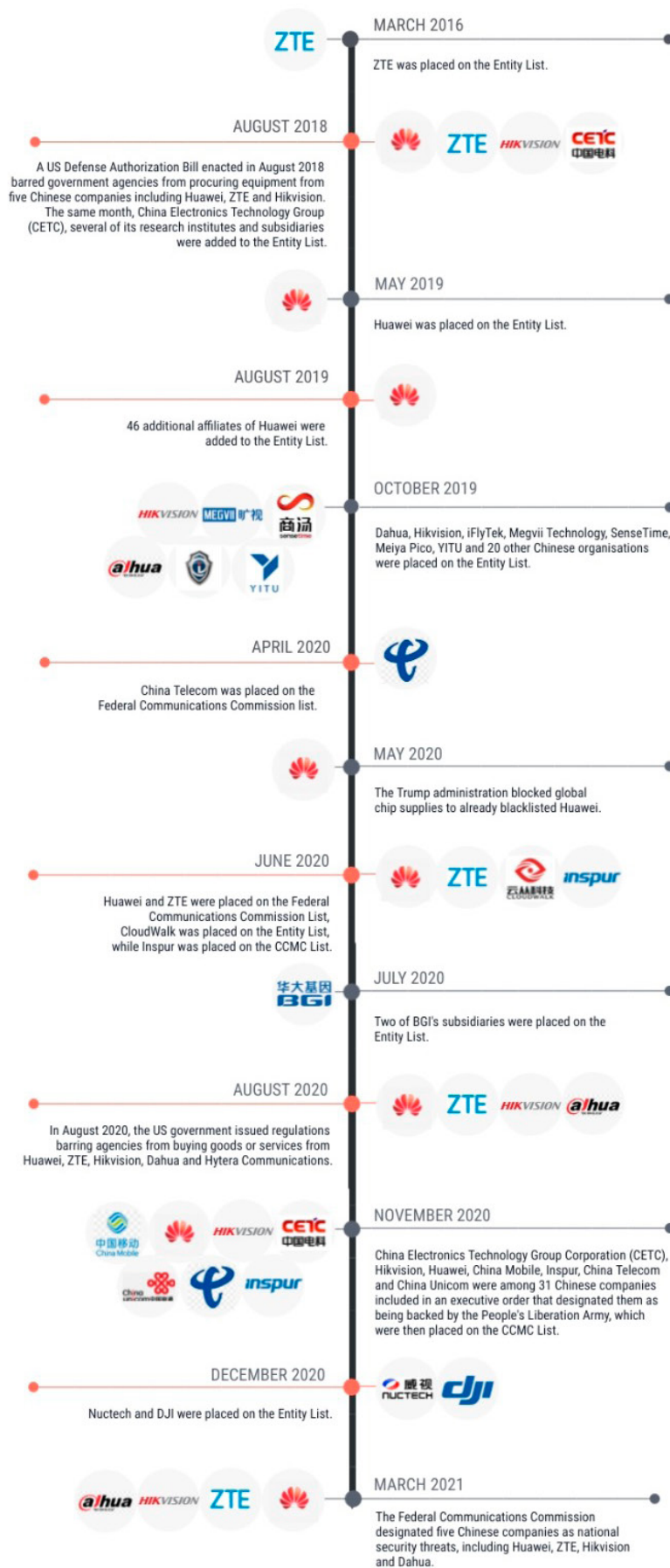
As factories in China were shut down and exports from the country ceased in China's early response to the Covid-19 outbreak, the pandemic triggered countries and companies to move away from their supply-chain reliance on China. Before the pandemic, the US Entity List played a role in the Trump administration's push to decouple the US economy from China. Cooperating with blacklisted companies on the Entity List raised fears among Western businesses about the data security and privacy risks associated with continued collaboration.⁶⁰ As those concerns and Entity List designations began affecting business between US and Chinese companies, the ramifications of the listings spread globally, influencing the actions of other countries against some of the technology giants on our map.

The impacts of the US Entity List and ensuing global actions against the Chinese technology companies that we observed have varied drastically, significantly slowing Huawei's overseas growth and overall expansion, while sparing major internet companies, including ByteDance, Tencent and Alibaba.⁶¹ The Entity List designation of telecommunications companies Huawei and ZTE prompted other countries, such as the members of the Five Eyes group and the EU, to implement policies aimed at limiting and in some cases excluding those companies from their 5G infrastructure. Although Covid-19 provided several surveillance and AI companies with an opportunity to neutralise such effects, many countries are still responding to security concerns associated with China's tech giants, and the impacts of further global actions can be expected to shift in severity in coming years.

In the five years since the US first blacklisted ZTE in 2016—in a move that threatened the corporate viability of the Chinese telecommunications company⁶²—Washington has widened its net to include a range of other Chinese companies, including 16 of the 27 featured on our map. As of April 2021, more than 400 Chinese companies, organisations and affiliates had been placed on the Entity List.⁶³

In addition to placing various Chinese companies on the Entity List, the Trump administration also prohibited US companies and citizens from investing in the securities of dozens of companies included in the Pentagon's list of 'communist Chinese military companies' operating in the US (the CCMC List),⁶⁴ including seven of the technology companies featured on our map: [China Electronics Technology Group \(CETC\)](#), [China Mobile](#), [China Telecom](#), [China Unicom](#), Hikvision, Huawei and [Inspur](#).⁶⁵ The Trump administration also proposed new rules that sought to eject Chinese firms from US stock exchanges for failure to comply with US auditing standards (Figure 5).

Figure 5: Timeline of US listings and other measures affecting Chinese tech companies



Note: For more information and sources, refer to Appendix 1.

3.1 The ZTE case

In March 2016, the US Department of Commerce added ZTE to the Entity List after it found that the company had schemed to hide its re-exports of US-origin items to Iran and North Korea, both of which were under US sanctions.⁶⁶ The restrictions prevented suppliers from providing ZTE with US equipment, threatening the company's supply chain.

While the ban brought the company to the brink of collapse, Washington extended a series of lifelines to ZTE, allowing it to maintain ties to its US suppliers before it agreed to pay US\$892 million in a plea deal in March 2017.⁶⁷ In April 2018, the US announced a seven-year ban on American firms selling parts and software to the company after it was found to be shipping US goods to Iran in violation of its agreement.⁶⁸

The ban had an immediate effect on ZTE, bringing the company's production to a grinding halt. It announced in April 2018 that it was ceasing 'major operating activities'.⁶⁹ The following month, US President Donald Trump threw an unexpected lifeline to the company, tweeting that there would be 'too many jobs in China lost' due to the US Government's actions against ZTE.⁷⁰

ZTE went on to report revenue growth hitting a five-year high during 2020. The company's operating revenue reached almost US\$16 billion (¥101.45 billion), indicating a year-on-year increase of 11.8%.⁷¹ Its net profit experienced a year-on-year increase of 17.3%, totalling US\$672 million (¥4.26 billion).⁷² While sales had declined in the US and Europe, the company was able to achieve sufficient growth in Asian markets and domestically, where it made over two-thirds of its revenue.

In August 2018, Washington reached for another tool. The annual Defense Authorization Bill barred government agencies from procuring equipment from five Chinese companies, including ZTE.⁷³ The Bill covered any substantial or essential technology component of any system used by US Government agencies, and especially mentioned technology used to track or view user data. As a result of the Bill, all agencies that were already using equipment provided by the Chinese companies were directed to allocate specific funding to replacing it.⁷⁴ When the Bill was enacted, it also targeted other Chinese companies, including Huawei and Hikvision.⁷⁵

3.2 Huawei's global struggles

Similarly to its competitor, ZTE, Huawei continues to experience turbulence due to its addition to the US's Entity List. The company was first blacklisted on 16 May 2019 by the US Commerce Department's Bureau of Industry and Security, together with 66 of its non-US affiliates.⁷⁶ The bureau later added several other affiliated entities in August 2019⁷⁷ and August 2020.⁷⁸

In addition to using the Entity List, the Trump administration blocked global chip supplies to Huawei in May 2020, further impeding the global expansion of the company's business.⁷⁹ As the crackdown on the company continued, Huawei was designated as a national security threat, together with ZTE, by the US Federal Communications Commission on 30 June 2020, which effectively barred them from receiving federal broadband subsidies to expand broadband access across the US.⁸⁰ Finally, in November 2020, Huawei and 30 other Chinese companies were included in an executive order that designated them as being backed by China's People's Liberation Army.⁸¹

As the US has taken action against Huawei, it has also actively encouraged and publicly pressured other countries to adopt similar policies.⁸² But many countries have taken their own, and often different pathways, to arrive at their decisions on 5G over the last few years. And some, like Australia, made their decisions long before the United States.

The Five Eyes countries have responded with some of the toughest policies against Huawei. In 2018, Australia became the first country to exclude ‘high-risk vendors’ from its 5G networks.⁸³ New Zealand similarly rejected Huawei’s first bid in the country in 2018 due to national security concerns.⁸⁴ The UK most recently banned mobile providers from purchasing new Huawei 5G equipment and announced that providers must remove all Huawei 5G equipment from their networks by 2027.⁸⁵ Although Canada hasn’t formally blocked Huawei, the country has delayed its decision long enough to effectively force its telecom companies to exclude Huawei equipment from their 5G networks.⁸⁶

According to the Dell’Oro Group, countries representing more than 60% of the world’s cellular-equipment market are now considering or have already acted to restrict Huawei.⁸⁷ The EU and several of its members have taken similar actions to block or limit Huawei’s presence in their 5G network deployments. In January 2020, the EU recommended that its members limit ‘high-risk 5G vendors’, including Huawei, stopping just short of recommending an outright ban of the company.⁸⁸ Swedish regulators banned wireless carriers from using Huawei’s 5G equipment, citing national security concerns. In response, however, Huawei challenged the decision in Swedish courts and has since threatened to exclude Ericsson from participating in China’s 5G growth.⁸⁹

Romania and Poland both enacted policies aimed at blocking Huawei from their 5G networks, although the policies didn’t explicitly ban Huawei.⁹⁰ Huawei sent a letter to the EU competition chief, in which the company argued that Poland’s and Romania’s proposed 5G security rules were ‘predicated on several violations of EU law’.⁹¹ In its letter, Huawei also cited the involvement of the US in those actions against the company, referencing ‘joint declarations’ and ‘memoranda of understanding’—aimed at pushing out 5G suppliers subject to foreign interference—that the US signed with several European countries, including Romania, Poland, Estonia, Latvia, the Czech Republic, Slovenia, Slovakia, Cyprus, Bulgaria, North Macedonia and Kosovo.⁹²

In 2020, as a result of global actions against it, Huawei reported its slowest annual revenue increase in a decade.⁹³ Specifically, Huawei’s revenue increased year-on-year by 3.8%, totalling US\$136.7 billion,⁹⁴ which was a drastic decline from its 19% revenue growth during 2019 (Table 1).⁹⁵ Although the company still managed to grow overall, China was the only region where it experienced positive revenue growth.⁹⁶ The company’s carrier business, which is responsible for building its telecom networks, grew by only 0.2%.⁹⁷ That stall was largely due to the decision of several Western countries to exclude Huawei’s 5G equipment from their networks.⁹⁸

Table 1: Huawei's 2020 business revenue, by region

	2020 (¥ million)	2019 (¥ million)	Change
China	584,910	506,733	15.4% ▲
Europe, Middle East and Africa	180,849	206,007	-12.2% ▼
Asia-Pacific	64,369	70,533	-8.7% ▼
Americas	39,638	52,478	-24.5% ▼
Other	21,602	23,082	-6.4% ▼
Total	891,368	858,833	3.8% ▲

Source: Huawei Investment & Holding Co. Ltd, 2020 annual report, 2021, [online](#).

While Huawei's decline in growth was most pronounced in North and South America in 2020, Europe, the Middle East and Africa collectively showed the next greatest decline, followed by the Asia-Pacific. This resulted in the company's decision to pivot its priority industries to focus on developing software. In an internal memo made public in May 2021, Huawei founder Ren Zhengfei wrote that Huawei should strive to 'lead the world' in software as the company seeks growth beyond its hardware operations.⁹⁹

Although Huawei's deputy chairman, Eric Xu Zhijun, said in an interview that the company's goal for 2021 is 'to survive', experts such as Dan Wang, an analyst with Gavekal, have speculated that Huawei may pivot to new businesses, such as self-driving and electric-vehicle technologies.¹⁰⁰ Already, Huawei reportedly has plans to invest US\$1 billion into researching self-driving and electric vehicles and is reportedly in talks to acquire a domestic automaker's electric vehicle unit.¹⁰¹ Through investing in businesses that are less reliant on advanced chips and through strengthening its software business, Huawei is searching for new revenue sources.¹⁰²

US sanctions have particularly affected Huawei's access to international technologies, such as advanced chips, that are essential for the company's products. When the US Government barred Huawei from purchasing semiconductors produced using US software or technology without a special licence, the move crippled Huawei's smartphone business and resulted in the sale of its Honor budget smartphone brand.¹⁰³ US sanctions also required Google to revoke Huawei's Android licence, leaving the company without access to Google apps and services that have been critical for the functioning of Huawei's smartphones.¹⁰⁴

In response to losing its Android licence, Huawei created a 'forked' version of Android to serve as its own operating system, Harmony OS, which is likely to face challenges as it seeks to attract developers and create apps.¹⁰⁵ If it's successful, however, Harmony OS would provide Huawei with complete control over an operating system with potential implementation in smartphones internationally, enabling Huawei to control the information environment—including which apps are banned—outside of China's borders.¹⁰⁶

Despite losing access to several markets globally, Huawei has signed new 5G and cloud-computing agreements with countries in Africa, the Middle East and Southeast Asia (Figure 6). Access to those markets will be critical for Huawei's future as the US and the EU move to confront their supply-chain dependence on China.¹⁰⁷

Figure 6: Huawei's 5G and cloud-related overseas presence



Note: The [Mapping China's Technology Giants](#) project website contains 200 data points of overseas presence relating to 5G and cloud technologies for Huawei.

3.3 Sanctions for all

Similarly to Huawei, state-controlled surveillance technology company Hikvision was added to the US's Entity List in October 2019.¹⁰⁸ Along with Hikvision, six other technology giants on our map were added at that time, including surveillance company *Dahua*, AI companies *iFlytek*, *Megvii*, *SenseTime*, and *YITU*, and digital forensics and security company *Meiya Pico*.¹⁰⁹

Although Hikvision's growth was boosted by Covid-19, a March 2020 disclosure detailed the negative impacts of sanctions on the company's overseas market and income. The disclosure stated that, as a result of its Entity List designation, Hikvision had increased its R&D costs significantly to allow for expanding upstream technology, changing materials and adjusting product designs.¹¹⁰ Additionally, Hikvision has been restricted in other countries, such as India, where the company is prohibited from bidding on government projects.¹¹¹ The company also faces scrutiny in Australia, where, as recently as January 2021, the South Australian health department removed all cameras made by Hikvision from public hospitals and nursing homes.¹¹²

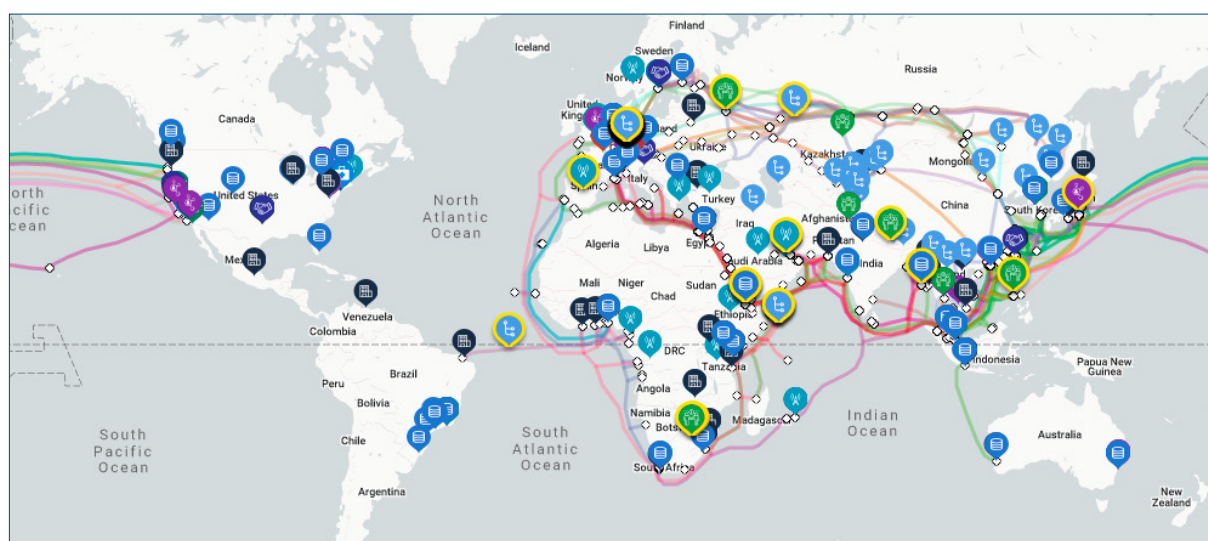
Predicting its addition to the Entity List in 2019, Hikvision stockpiled essential components in preparation, which proved helpful in mitigating the immediate impacts.¹¹³ As the global chip shortage continues to affect the technology industry, however, Hikvision's president has indicated future uncertainties for the company if the situation persists.¹¹⁴

Among the companies we tracked, BGI Group—a key supplier of Covid-19 testing technology—experienced the greatest growth despite being blacklisted by the US. In July 2020, the US Department of Commerce placed two of BGI's subsidiaries (*Xinjiang Silk Road BGI* and *Beijing Liuhe BGI*) on the Entity List.¹¹⁵ However, due to the company's key role in providing Covid-19 testing equipment, BGI reported a surge in its net profit and share price during 2020.

Other Chinese tech companies on our map that were affected by US sanctions include DJI and [Nuctech](#). The US Department of Defense first issued a ban on the purchase and use of DJI's commercial drones on 23 May 2018 and later added the company to the export blacklist in December 2020.¹¹⁶ Although DJI continued to expand during 2020, it faced challenges in maintaining its large presence overseas, reportedly having to make sweeping cuts to its global sales and marketing teams.¹¹⁷ Despite its Entity List designation, DJI maintains control of more than 70% of the global drone market, and North America remains its largest market.¹¹⁸

China's major telecommunications companies—China Telecom, China Unicom and China Mobile—have been targeted by Washington in several capacities (Figure 7). Most recently, in January 2021, the three companies were added to the Pentagon's CCMC List, which triggered a series of delistings and relistings of the companies by the New York Stock Exchange, eventually resulting in the final delisting of all three.¹¹⁹ The companies were also among 31 Chinese companies included in a November 2020 executive order that designated them as being backed by the People's Liberation Army.¹²⁰ Before those designations, the US Federal Communications Commission had already begun taking action against China Telecom and China Unicom in April 2020.¹²¹ Despite being added to the lists, all three telecom companies experienced growth during 2020 as they expanded their 5G operations—especially in China.

Figure 7: Chinese telcos' overseas presence



Note: The [Mapping China's Technology Giants](#) project counts more than 480 points of overseas presence for China's three major telecommunications operators (China Mobile, China Telecom and China Unicom) combined.

Apart from those tech giants, several major Chinese technology companies on our map have been largely spared US economic countermeasures, specifically Alibaba, Ant Group, Baidu, ByteDance and Tencent. There were, however, disparate attempts by the Trump administration to take action against those companies, which all eventually failed during Trump's term of office.

In January 2021, for instance, the US Department of State and Department of Defense pushed to add Alibaba, Tencent and Baidu to the CCMC List, which would have banned US investors from holding stock in the three companies.¹²² Previously, in August 2020, Trump issued two executive orders prohibiting any American company or person from conducting transactions with ByteDance, which is TikTok's parent company, and Tencent's WeChat.¹²³ The bans were halted a month later by a US

federal judge, citing First Amendment rights.¹²⁴ In October 2020, the US State Department proposed adding Ant Group to the Entity List, which was seen as a move to discourage US investors from taking part in Ant's upcoming IPO in Shanghai and Hong Kong. The bid was later put on hold by the Trump administration.¹²⁵ Any impacts of attempted bans on those companies were neutralised as demand for digital products skyrocketed during the Covid-19 pandemic.

Although the attempts to take action against Alibaba, Ant Group, Baidu, ByteDance and Tencent were unsuccessful, they attracted global attention to the data privacy and security risks associated with using products and applications developed by the Chinese technology giants. Following US attempts, India permanently banned 59 Chinese apps from its domestic market in January 2021, while Germany's intelligence agencies warned consumers that personal data provided to Chinese technology companies could end up in the possession of the Chinese Government.¹²⁶ As the US and other countries continue targeting China's tech giants through various regulatory measures, they're being pushed to address their reliance on China just as China is seeking to reduce its dependence on the US for critical technologies, particularly semiconductors.

4. Localising supply chains: from a ‘choke point’ to ‘dual circulation’

From the perspective of Beijing’s policymakers, 2020 was a year in which, as Vice Foreign Minister Le Yucheng put it, China experienced a ‘plot reversal’ and ‘turned a crisis into an opportunity’.¹²⁷ ‘Rather than being a “Chernobyl moment” for China, the pandemic became a ‘highlight moment for socialism with Chinese characteristics’, Le told a think-tank forum in December 2020. The triumphalist note came as China’s ability to contain the spread of Covid-19 before other major economies allowed it to rebound faster and end 2020 on a high note as the only major economy to report positive growth, achieving an economic expansion of 2.3%.¹²⁸

Despite their upbeat tone, China’s leaders also recognised that the combination of the Covid-19 pandemic and the US–China trade war had exposed the country’s fragility in technological innovation. In a speech to scientists in September 2020, Xi Jinping stressed the need for China to ensure secure and stable supply chains and to pursue indigenous innovation: ‘We must give full play to the significant advantages of our country’s socialist system that concentrate power on large undertakings, and successfully fight tough battles for the key core technologies,’ he instructed.¹²⁹

While the Chinese state’s goal of achieving self-reliance in technology has been a longstanding policy, the combination of the Covid-19 pandemic and the ever-tightening technology blockade imposed by the White House put the issue front and centre for the Chinese leadership. In December 2020, China’s Central Economic Working Conference announced that science and technology work would be the top priority in 2021. The 14th Five-Year Plan, unveiled in March 2021, described technological innovation as a matter of national security, not just economic development, for the first time.¹³⁰

4.1 Mobilising the tech industry

China’s technology companies are set to play a key role in addressing that fragility as they’re mobilised in what Beijing’s top policy official, Jiang Jinqian, calls a ‘whole country approach’ to reduce reliance on foreign technologies.¹³¹ That effort would seek breakthroughs in ‘strategic and fundamental key science and technology projects’ so that the country can overcome ‘choke points’ in its technological progression, Jiang said in his interpretation of an as yet unpublished keynote speech made by Xi to China’s provincial-level leaders in early January 2021. As part of the plan, the country will establish ‘national teams’ to strengthen scientific research and innovation, according to Jiang. The private sector will be encouraged to invest in R&D, and the state will reward companies through ‘state purchase of research results’.

Several of the companies featured on our map, including SenseTime, Huawei, ZTE, Megvii, YITU, CloudWalk, Baidu, Alibaba, Tencent and China’s three major telcos, have already been recruited in a US\$2 trillion new infrastructure campaign that the Chinese state introduced in the early days of the pandemic to boost the economy and cushion the impact of the global slowdown. The campaign targets high-tech sectors such as 5G infrastructure, AI, big data centres, the industrial internet, ultra-high-voltage high-speed intercity rail and electric vehicle charging infrastructure.¹³² The plan is largely a continuation of the Made in China 2025 campaign that was launched in 2015, with some minor cosmetic changes.

Made in China 2025 targeted investments in 10 strategic industries now largely dominated by the US, including aerospace, semiconductors, information technology, robotics, green energy, electric vehicles, agricultural machinery, pharmaceuticals and advanced materials. The campaign attracted sustained criticism from the Trump administration for its attempt to capture market share from China's foreign technology rivals. The new infrastructure campaign dropped any reference to that plan as well as any explicit requirements that core technology must be sourced domestically. The campaign is funded mainly by the private sector and local governments instead of the national government.¹³³

China's three national telecom carriers (China Unicom, China Telecom and China Mobile) collectively promised in March 2020 to invest around US\$34 billion (¥220 billion) to build 5G base stations in China. Tencent said that it would invest US\$77 billion (¥500 billion) over the following five years in new infrastructure technologies, such as cloud computing, and cybersecurity. Alibaba also pledged US\$30 billion (¥200 billion) in new infrastructure investments over three years.

4.2 All about the chips

Over the long term, the success of the new infrastructure campaign hinges on China's access to the world's most advanced semiconductor chips, which are the basic building blocks for emerging technologies such as 5G, AI and autonomous vehicles, in which Beijing hopes to lead the world. China's reliance on a globalised value chain to source semiconductor chips is seen by Chinese leaders from Xi Jinping down as a key obstacle to the country's technological ambitions.

The Trump administration's assault on China's ability to source semiconductor chips resulted in a flurry of panic buying. Imports of semiconductors jumped by 33.6% to US\$155.6 billion in the first three months of 2021—an increase of 77.6% from 2019.¹³⁴ Beijing's attempts at achieving self-sufficiency in semiconductors have been beset by setbacks, and large subsidies for semiconductor projects have failed to produce successes. China's self-sufficiency ratio for semiconductors is expected to be only 19.4% in 2025.¹³⁵

In an effort to achieve self-sufficiency, public and private entities in China have facilitated the organisation of several technology-focused alliances. In 2016, Huawei, ZTE, Inspur and the Ministry of Industry and Information Technology were among 27 entities that established China's High End Chip Alliance, which aims to promote the production of, research into and collaborative innovation on chip technology.¹³⁶ The National Integrated Circuit Standardisation Technical Committee was later proposed by the China Electronics Standardisation Institute in 2021. Huawei, Tencent and Alibaba are among 90 Chinese tech companies that joined the committee in an effort to strengthen the domestic semiconductor supply chain.¹³⁷

Huawei's addition to the US's Entity List further spurred its efforts to create a domestic supply chain but it also served as a warning to other Chinese tech companies featured on our map, such as ByteDance, Baidu, Alibaba and SenseTime, that now view reliance on US technology as a vulnerability that must be eliminated. ByteDance is exploring the feasibility of developing its own AI chips.¹³⁸ Baidu has completed one round of financing for its Kunlun AI chip unit and is considering commercialising its chip design capabilities.¹³⁹ Alibaba has also unveiled an AI chip for its cloud-computing products.¹⁴⁰ After being added to the Entity List in 2019, SenseTime began developing its own AI chips.¹⁴¹ Meanwhile, Huawei is reportedly constructing a dedicated chip plant in Shanghai that won't use American technology.¹⁴²

4.3 Dual circulation

The Covid-19 pandemic and the growing China–US strategic and technological competition also prompted a major rethink in economic policy for the CCP. A new strategy began to take shape in a series of key speeches and party documents as China emerged from its Covid-19 economic slump in early 2020. In April 2020, in a seminal speech on China’s economic development that was kept under wraps for six months, Xi Jinping said that the impact of the pandemic had exposed hidden risks in China’s industrial and supply chains and that the country ‘must strive to have at least one alternative source for key products and supply channels, to create a necessary industrial backup system’.¹⁴³

Referred to as a need to speed up China’s ‘dual circulation’ growth model, the new economic strategy became the focus of the 14th Five-Year Plan adopted on 11 March 2021, which charts a course for China’s economy from 2021 to 2025.¹⁴⁴ It envisages a future in which Beijing steadily weans itself off high-end imports from industrialised nations while using the ‘powerful gravitational field’ of its economy to make other nations heavily reliant on China for high-tech supplies and as a market for raw materials. As Xi said in his April 2020 speech:

We must sustain and enhance our superiority across the entire production chain ... and we must tighten international production chains’ dependence on China, forming a powerful countermeasure and deterrent capability against foreigners who would artificially cut off supply [to China].

By pursuing a strategy of ‘dual circulation’, Beijing hopes to build fully domestic supply chains while binding foreign companies to the Chinese market even more strongly. Over the long term, the aim is for a stronger China able to withstand economic coercion, but also for China to be in a stronger position to inflict coercion on other countries. The CCP’s use of economic coercion against countries such as Australia and companies such as Swedish retailer H&M foreshadow how the Chinese state is likely to use its enhanced power if its ‘dual circulation’ strategy is successful.

5. Reining in the tech giants: tougher regulation at home

China's regulatory agencies have treated the country's tech giants with a light touch for most of the companies' history, favouring their pursuit of technological dominance and economic prosperity over the need for regulating their growing monopoly power.

In October 2020, the scales tipped in the opposite direction after Jack Ma, the co-founder of Alibaba and its fintech affiliate, Ant Group, made a public speech in Shanghai in which he levelled a scathing critique of financial regulators and implicitly rejected Xi Jinping's signature campaign to combat financial risks.¹⁴⁵ The speech reportedly infuriated the leadership in Beijing and prompted Xi to personally call off Ant Group's impending US\$34 billion IPO and order regulators to investigate risks posed by Ma's business.¹⁴⁶

Regulators cited the systemic financial risks posed by Ant Group as the reason for the company to reorganise itself as a financial institution, subject to oversight by the country's central bank, the People's Bank of China. Escalating geopolitical tensions with the US and the ensuing US–China trade war contributed to the regulator's efforts to rein in Ant Group, as Beijing sought to head off risks in the banking system amid concerns that the stand-off with Washington could precipitate a financial crisis.

Ma's speech served as a tipping point for agencies, such as China's antitrust authority, the State Administration for Market Regulation (SAMR), that have now become much more assertive with their agenda to draw clear lines between tech companies and financial services companies—lines that Jack Ma was intending to further blur. As Ma removed himself from public view, the campaign widened out to other companies in late April 2020, when the People's Bank of China and four other regulatory agencies told 13 firms, including Tencent and ByteDance, that their apps should no longer provide financial services beyond payments.¹⁴⁷

Ma's speech may have been a catalyst for some regulatory agencies, but the groundwork for action had been put in place much earlier. In January 2020, the SAMR proposed the first major revisions to the country's 2008 antimonopoly law in over a decade, including provisions for large internet platforms.¹⁴⁸ The regulatory push has been spearheaded by Vice Premier Liu He, who is Xi Jinping's top economic adviser.¹⁴⁹ The principles underlying the campaign—'tackling monopolies' and 'preventing disordered capital expansion'—emerged during several high-level government meetings, including the Fifth Plenary Session of the 19th CCP Central Committee in October 2020 and the Central Economic Working Conference at the end of the year.¹⁵⁰

Beijing's effort to tame the outsized power of China's internet companies has continued to widen. A week after Ant Group's IPO was scuttled, the SAMR published draft rules to curb monopolistic behaviour in the country's tech sector, immediately wiping US\$280 billion from the market capitalisation of the internet giants Tencent, Xiaomi, Meituan and JD.com.¹⁵¹ In April 2021, Alibaba Group was hit with a record US\$2.81 billion antimonopoly fine, which was equivalent to around 4% of the group's 2019 revenue. An investigation into Tencent is currently underway, and some reports suggest that it, too, may be hit with a fine of at least US\$1.54 billion (¥10 billion).¹⁵²

The SAMR went on to summon 34 technology companies and warn them to ‘heed the warning’ provided by Alibaba’s case. The companies, which included Baidu, Tencent and ByteDance, were given one month to undergo ‘complete rectification’ to ensure that they weren’t in breach of anti-monopoly laws. In a statement, the monopolies regulator stressed that the companies must ensure that they’re not doing anything that ‘harms the interests of operators and consumers’ and that they should give ‘priority to national interests’.¹⁵³ Between December 2020 and April 2021, the regulator fined 11 companies, including Tencent, Baidu, Alibaba and ByteDance, for failing to disclose past acquisitions and investments.¹⁵⁴ As the government continues to clamp down on this sector, investors have grown nervous, leading to a plunge in the combined market capitalisation of 10 leading technology companies by over US\$800 billion from its peak in February 2021.¹⁵⁵

Beijing’s campaign, which is set to continue throughout 2021, comes at the same time as efforts in the West to rein in companies such as Facebook and Google have gained momentum. The efforts share some similar worries: regulators in the US, Europe and China all cite concerns that the technology giants have built market power that stifles competition, misuses consumer data and violates consumer rights. But, for China’s regulators, the need to discipline their country’s tech companies goes beyond those concerns to a broader sense that the companies’ interests aren’t sufficiently lined up with the CCP’s industrial policy or its goal of achieving technological self-sufficiency.

An editorial in the *People’s Daily* in December 2020 urged the country’s internet giants to focus on innovation instead of the ‘community group-buying’ market.¹⁵⁶ ‘Internet giants with access to big data and advanced computing should have a greater responsibility, greater pursuits, and a greater role in scientific and technological innovation,’ the CCP mouthpiece wrote. The CCP has now moved on from merely chiding the tech companies to enforcing their adherence to its strategic goals. In January 2021, the head of the SAMR emphasised that one of his priorities for 2021 was to ‘promote the coordination of industrial policy and competition policy’.¹⁵⁷

6. Conclusion

The Covid-19 pandemic may have been a short-term boon to many of China's technology giants, but, for the CCP, the pandemic and the US–China trade war were a stark reminder of the country's fragility in technological innovation. While the Chinese state's goal of achieving self-reliance in technology has been a longstanding policy, the combination of the Covid-19 pandemic and the ever-tightening technology blockade imposed by the White House elevated the issue to a higher level of importance than ever before.

The onslaught of sanctions and other related measures from the US helped to further align the interests of China's tech giants with the CCP's goal of achieving technological self-sufficiency. A newly launched rectification campaign in the technology sector is designed to ensure that this alignment continues. The campaign, which looks set to continue throughout 2021 and beyond, is already bearing fruit as major internet companies warn investors that they're preparing to funnel capital into areas that the Chinese state has identified as priorities, such as cloud computing, autonomous vehicles and AI.¹⁵⁸

Already, a string of high-level resignations have taken place in various Chinese technology companies, including Ant Group, Pinduoduo and ByteDance, as the government seeks to weaken the central authority of all the leaders of the major technology companies.¹⁵⁹ The Chinese state is embarking on a fundamental restructuring of the technology industry and the private sector more broadly so that, as CCP guidelines released in September 2020 put it, 'ideological guidance' is strengthened to 'create a core group of private sector leaders who can be relied upon during critical times'.¹⁶⁰

The Chinese state is more determined than ever to rein in China's technology giants and push them, and the country, towards technological self-sufficiency.

Appendix 1: Timeline of US entity listings and other measures

2014	The Transportation Security Administration effectively banned Nuctech from the American airport security market. ¹⁶¹
8 March 2016	The Department of Commerce added ZTE to the Entity List after it found that the company had schemed to hide its re-exports of US-produced items to Iran and North Korea, which were under US sanctions. ¹⁶²
29 March 2017	The Department of Commerce removed ZTE Corporation and ZTE Kangxun Telecommunications Ltd from the Entity List after the US Government 'reached an agreement with ZTE Corporation and ZTE Kangxun for the settlement of administrative charges and entry of a guilty plea in a criminal case against the companies'. ¹⁶³
April 2018	The Department of Commerce banned American companies from selling components to ZTE for seven years after finding that ZTE had broken an agreement that forbade it to ship goods to Iran. ¹⁶⁴
23 May 2018	The Department of Defense issued a ban on the purchase and use of DJI's 'commercial over-the-shelf unmanned aerial systems' because of cybersecurity risks. ¹⁶⁵
July 2018	The Department of Commerce lifted the April 2018 ban against ZTE after the company paid a US\$1.4 billion fine. ¹⁶⁶
1 August 2018	China Electronics Technology Group (CETC) and several of its research institutes and subsidiaries were added to the Entity List due to being 'involved in the illicit procurement of commodities and technologies for unauthorized military end-use in China'. ¹⁶⁷
13 August 2018	The Defense Authorization Bill enacted in August 2018 barred government agencies from procuring equipment from five Chinese companies, including Huawei, ZTE and Hikvision. ¹⁶⁸
16 May 2019	The Department of Commerce's Bureau of Industry and Security added Huawei and 66 of its non-US affiliates to the Entity List. ¹⁶⁹
19 August 2019	Forty-six additional Huawei affiliates were added to the Entity List. ¹⁷⁰
9 October 2019	The Department of Commerce placed Dahua, Hikvision, iFlytek, Megvii Technology, SenseTime, Meiya Pico, YITU and 20 other Chinese organisations on the Entity List. ¹⁷¹
April 2020	The Federal Communications Commission initiated proceedings to revoke and terminate China Telecom (Americas) Corporation's domestic and international authorisations. ¹⁷²
May 2020	The Trump administration blocked global chip supplies to already blacklisted Huawei. ¹⁷³
5 June 2020	The Department of Commerce placed CloudWalk on the Entity List for 'engaging in activities contrary to the foreign policy interests of the United States'. ¹⁷⁴
12 June 2020	Inspur Group was added to the US Department of Defense's CCMC List 'in accordance with the statutory requirement of Section 1237' of the fiscal year 1999 National Defense Authorization Act. ¹⁷⁵
30 June 2020	The Federal Communications Commission designated ZTE and Huawei as national security threats, effectively barring them from receiving federal broadband subsidies to expand broadband access across the US. ¹⁷⁶
22 July 2020	The Department of Commerce placed two of BGI's subsidiaries (Xinjiang Silk Road BGI and Beijing Liuhe BGI) and nine other Chinese organisations on the Entity List. ¹⁷⁷
8 August 2020	President Trump issued executive orders prohibiting any American company or person from conducting transactions with ByteDance, which is TikTok's parent company, and Tencent's WeChat. This was later halted by a US federal judge. ¹⁷⁸
August 2020	The US Government issued regulations barring agencies from buying goods or services from Huawei (adding 28 extra non-US affiliates ¹⁷⁹), ZTE, Hikvision, Dahua and Hytera Communications. ¹⁸⁰
October 2020	The State Department proposed adding Ant Group to the Entity List, which was seen as a move to discourage US investors from taking part in Ant's upcoming IPO in Shanghai and Hong Kong. The bid was later put on hold by the Trump administration. ¹⁸¹

November 2020	CETC, Hikvision, Huawei, China Mobile, Inspur, China Telecom and China Unicom were among 31 Chinese companies included in an executive order that designated them as being backed by the People's Liberation Army and were added to the CCMC List. ¹⁸²
December 2020	The Trump administration added Chinese semiconductor and drone manufacturers to the export blacklist, including DJI and Nuctech. ¹⁸³
5 January 2021	President Trump issued an executive order prohibiting any transaction between American people or other entities and eight Chinese apps, including Ant group's Alipay and Tencent's Tencent QQ and WeChat Pay. ¹⁸⁴ The Biden administration has since distanced itself from attempts by the previous administration to ban WeChat. ¹⁸⁵
6 January 2021	The New York Stock Exchange announced that it would move forward with the delisting of China Telecom, China Mobile and China Unicom. ¹⁸⁶
7 January 2021	The Department of Defense and Department of State pushed hard to add Alibaba, Tencent and Baidu to the CCMC List, which would have banned US investors from holding stock in the three companies. ¹⁸⁷ Treasury Secretary Steven Mnuchin argued that putting Alibaba on the list would create a huge problem for US investors. As a result, the Treasury Department blocked the attempt.
17 March 2021	The Federal Communications Commission designated five Chinese companies as national security threats, including Huawei, ZTE, Hikvision and Dahua. ¹⁸⁸
17 March 2021	The Federal Communications Commission launched a process to determine 'whether to end China Unicom Americas' authority to provide domestic interstate and international telecommunications services within the United States'. It cited vulnerabilities to exploitation, influence and control by the Chinese Government. ¹⁸⁹
May 2021	China Mobile, China Unicom and China Telecom announced that they would be delisted by the New York Stock Exchange, after their appeal to reverse the move failed. ¹⁹⁰
3 June 2021	US President Biden signed an Executive Order to amend EO 13959 'Addressing the Threat from Securities Investments That Finance Communist Chinese Military Companies'. Companies previously designated as 'Communist Chinese military companies' by the US Department of Defense under Trump remained on Biden's EO, including CETC, China Mobile, China Telecom, China Unicom, Hikvision, Huawei, and Inspur. All 59 entities included in this amendment to EO 13959 were also added to the US Department of Treasury's new Non-SDN Chinese Military-Industrial Complex Companies List (NS-CMIC List). ¹⁹¹

Acronyms and abbreviations

AI	artificial intelligence
CCMC List	Communist Chinese Military Companies List
CETC	China Electronics Technology Group Corporation
CCP	Chinese Communist Party
EU	European Union
GDP	gross domestic product
IPO	initial public offering
R&D	research and development
SAMR	State Administration for Market Regulation (China)
UK	United Kingdom

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